



FINAL YEAR DESIGN PROJECT (FYDP-II) • FINAL DEFENSE

NeuroVerse

An AI-Integrated Student Mental Health & Wellness Ecosystem with Decoupled Real-Time Microservice Intelligence

Course: CSE 4800 / CSE 4889 Final Year Design Project II

Department: Department of Computer Science & Engineering

Institution: United International University (UIU)

Team: Team NeuroLink

Defense Date: Academic Year 2026

Status: Completed & Operational System

Team NeuroLink & Role Allocations

Multidisciplinary student team delivering the end-to-end full stack, machine learning & clinical flows



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- System Architecture
- Node/Express Backend API
- JWT Authentication & RBAC
- Cloud DB Management



MD Annan
ID: 011222128

- ML Pipeline Engineering
- NLP Sentiment Classification
- Crisis Triage Detection
- Model Benchmarking



Wasiur Rahman Sakib
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- FastAPI ML Microservice
- Target Variable Engineering
- Data Leakage Resolution
- Linear Regression Models



Md. Asif Sarkar
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- React 18 Frontend UI
- Wellness Dashboard Widgets
- Interactive Data Visuals
- Axios Client Integration



Mirza Samawatun Astha
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- Anonymous Peer Community
- Therapist Booking System
- Clinical Safety Flows
- User Experience & Testing

The Student Mental Health Crisis

Academic pressures and severe systemic barriers prevent university students from seeking timely help

73.5%

ACADEMIC STRESS RATE

Academic Pressure

Intense competition for CGPA, relentless exam deadlines, and uncertain career prospects cause high chronic stress and burnout among university students.

68.2%

UNREPORTED DUE TO STIGMA

Social Stigma & Fear

Students fear being labeled or judged by peers, faculty, and family. On-campus counseling centers lack privacy, deterring students from visiting.

1:5000+

COUNSELOR-STUDENT RATIO

Resource Scarcity

Campus mental health services are critically understaffed, leading to multi-week waiting queues that leave students in crisis without urgent care.

4+ Apps

DISCONNECTED TOOLS

Tool Fragmentation

Students must juggle separate apps for journaling, meditation, and therapy. Existing global apps ignore the Bangladeshi student context and require paid subscriptions.

The Implemented Solution: NeuroVerse Ecosystem

A unified, secure, AI-powered digital wellness ecosystem built specifically for university students

1. Proactive AI Intelligence

- **Demographic Mood Regressor:** Calculates continuous baseline wellness (1.0–5.0).
- **NLP Crisis Triage:** Scans journal reflections for suicide/self-harm triggers.
- **Knowledge Graph:** Generates tailored self-care learning paths.

2. Anonymous Peer Support

- **Zero Identity Leakage:** Pseudonymous posting protects student privacy completely.
- **Topic Categorization:** Exam anxiety, loneliness, relationship stress, burnout.
- **Automated Moderation:** Real-time filtering shields students from toxicity.

3. Verified Clinical Access

- **Direct Scheduling:** Browse accredited psychologists with verified credentials.
- **Flexible Booking:** Seamless slot reservation tailored to class schedules.
- **Clinical Confidentiality:** Encrypted communication and isolated therapy notes.

4. 24/7 Self-Help Hub

- **Immediate Emergency Hotlines:** Instant access to national crisis helplines.
- **Interactive Audio Tools:** Guided box-breathing & stress relaxation audio.
- **Psycho-education:** Evidence-backed student cognitive coping strategies.

UN Sustainable Development Goals (SDG) Alignment

NeuroVerse directly advances three core United Nations 2030 Sustainable Development Goals

TARGET 3.4 & 3.5

SDG 3: Good Health & Well-being

- Promotes mental health and reduces premature mortality through early intervention.
- Sub-15ms NLP crisis detection identifies suicidal ideation and connects students with emergency helplines before escalation.
- Continuous mood telemetry prevents severe depressive episodes.

TARGET 4.7

SDG 4: Quality Education

- Alleviates academic burnout and cognitive fatigue, improving student retention and performance.
- Provides evidence-based coping mechanisms for exam distress and time management.
- Creates emotionally supportive digital campus ecosystems conducive to lifelong learning.

TARGET 10.2

SDG 10: Reduced Inequalities

- Democratizes high-tier mental healthcare regardless of socioeconomic background or campus tier.
- 100% free core services eliminate financial barriers inherent in commercial Western therapy apps.
- Anonymous peer support bridges the social stigma gap for marginalized and vulnerable students.

Benchmark Analysis: Existing Solutions vs NeuroVerse

NeuroVerse uniquely addresses the complete student mental health lifecycle within a single platform

Feature / Capability	MindDoc	Headspace	7 Cups	BetterHelp	NeuroVerse (Ours)
Mood & Lifestyle Tracking	✓ Advanced	X None	X Basic	X None	✓ Full Multi-Metric
Anonymous Campus Peer Community	X None	X None	✓ Global Only	X None	✓ Student Dedicated
Real-Time NLP Crisis Triage (Holdout 84.6%)	X Questionnaire Only	X None	X Volunteer Triage	X Intake Survey	✓ Automated ML (<15ms)
Demographic Stress Regression	X None	X None	X None	X None	✓ Linear Baseline Model
Verified Therapist Scheduling	X Directory Only	X None	X External	✓ Paid (\$260+/mo)	✓ Built-in Scheduling
Student Focused & Free Core Access	X Freemium	X Paid (\$70/yr)	✓ Partial	X Expensive	✓ 100% Free Core Tools

Methodology: Agile Scrum Execution Across FYDP-I & II

Structured two-week iterative sprints transitioning theoretical design into an operational platform

1. Scrum Framework & Ceremonies

- **Two-Week Sprints:** Focused iterative cycles driving feature delivery and risk mitigation.
- **Daily Stand-ups:** Rapid 15-minute syncs identifying technical blockers and dependency handoffs.
- **Sprint Reviews & Retros:** Bi-weekly supervisor demonstrations and architecture fine-tuning.
- **Scrum Roles:** Clear role separation across Full-Stack, ML Microservice, and Quality Assurance.

2. Phased Delivery Roadmap

- **FYDP-I Phase (Foundation):**
 - Student surveys & requirement elicitation
 - Figma UI/UX prototyping
 - Initial MERN architecture setup & MongoDB schemas
- **FYDP-II Phase (Execution & ML):**
 - Decoupled FastAPI microservice implementation
 - 5 ML task pipelines & leakage resolution
 - Full-stack integration & API performance tuning

3. Quality Gates & Verification

- **Strict Definition of Done (DoD):** Every route backed by unit tests, RBAC checks, and documentation.
- **Continuous Integration:** Code reviews on GitHub before merging into staging branch.
- **Empirical Validation:** Machine learning evaluation on dedicated 20% holdout test sets.
- **Performance Gates:** All API routes profiled under concurrent loads to guarantee sub-25ms latency.

System Architecture: Three-Tier Decoupled Microservice

Isolating data-intensive machine learning inference from high-concurrency web transactions

TIER 1 • CLIENT

Presentation Layer

- **React 18 & Vite:** Ultra-fast client-side SPA rendering with modular components.
- **Tailwind CSS:** Highly responsive, accessible, dark/light unified UI.
- **Axios REST Client:** Interceptors managing JWT bearer cookies and error handling.

TIER 2 • PRIMARY API

Business Logic Server

- **Node.js & Express.js:** High-throughput, event-driven HTTP API gateway.
- **JWT & HTTP-Only Cookies:** State-of-the-art authentication shielding XSS exploits.
- **Role-Based Access Control:** Strict Student / Therapist / Admin authorization guards.

TIER 3 • ML ENGINE

FastAPI Microservice

- **Python 3.10 & Uvicorn:** Asynchronous ASGI microservice dedicated to ML inference.
- **Scikit-Learn & NLTK:** TF-IDF vectorizers, linear regressors, logistic classifiers.
- **NetworkX & Embeddings:** SentenceTransformers graph traversals for learning paths.

TIER 4 • PERSISTENCE

MongoDB Atlas Cloud

- **MongoDB Atlas (v7.0):** Globally distributed, auto-scaling NoSQL document database.
- **Mongoose ODM:** Schema enforcement, compound indexing, encrypted sensitive fields.
- **Automated Backups:** Point-in-time recovery and 99.99% uptime guarantee.

Architectural Deep-Dive: Why Decouple the ML Service?

Protecting the Node.js single-threaded event loop from CPU-bound mathematical bottlenecks

⚠ The Monolithic / Coupled Risk

Node.js uses a **single-threaded event loop** to process incoming requests concurrently.

- **CPU-Bound Blocking:** Matrix multiplications, TF-IDF n-gram vectorization, and graph DFS traversals consume 100% of thread execution.
- **Catastrophic Lag:** While calculating a single journal sentiment prediction, all other student requests (auth, page loads, forum browsing) freeze.
- **System Vulnerability:** A sudden influx of journal submissions during exam week could crash the entire university server.

✓ The Decoupled Hybrid Architecture (Implemented)

The Python **FastAPI microservice** operates in complete process isolation with worker pooling.

- **Non-Blocking Gateway:** Node.js delegates analytical calls asynchronously over internal HTTP and returns to serving user traffic instantly.
- **Independent Scalability:** The ML service can be vertically or horizontally scaled with multi-core Uvicorn workers without touching the web server.
- **Empirical Performance:** Microservice ML responses resolve in **1.2 ms** to **4 ms**, completely shielding users from latency spikes.

Backend Security, RBAC & Authentication Architecture

Enterprise-grade security controls safeguarding vulnerable student psychological telemetry

1. Cryptographic Hashing

- **Salted Bcrypt (Cost 10):** All user passwords undergo one-way cryptographic hashing before MongoDB persistence.
- **Zero Plaintext Exposure:** Passwords are never logged, transmitted over internal bridges, or returned in API payloads.
- **Timing-Attack Immunity:** Resistant against dictionary and rainbow-table attacks.

2. JWT in HTTP-Only Cookies

- **XSS Attack Immunity:** Tokens are transported inside strictly configured HTTP-Only, SameSite cookies.
- **Client Script Isolation:** Malicious browser scripts cannot read or exfiltrate session tokens from local storage.
- **Stateless Verification:** Backend verifies cryptographic HMAC signature on every protected route.

3. Role-Based Access Guards

- **Student Guard:** Grants access to personal wellness logs, anonymous forums, and booking requests.
- **Therapist Guard:** Restricts patient session logs, schedule slot creation, and clinical notes.
- **Admin Moderation Guard:** Provides system health metrics, user moderation, and emergency flag escalation.

Machine Learning Pipeline: 5 Core Intelligent Tasks & Datasets

Multimodal intelligence trained across five distinct empirical datasets to power comprehensive care

TASK 1 • REGRESSION

Demographic Mood Score

- **Dataset:** Student Mental Health Survey (101 samples)
- **Predictors:** Age, Gender, Year, CGPA, Marital Status
- **Model:** Linear Regression ($R^2 = 0.423$, MAE = 0.727)
- **Target:** Baseline score (1.0 to 5.0)

TASK 2 • NLP TRIAGE

Journal Sentiment & Crisis

- **Dataset:** Amalgamated Corpus (81,020 statements)
- **Features:** TF-IDF unigrams & bigrams (5K features)
- **Model:** Logistic Regression (81.8% acc, 84.6% crisis recall)
- **Latency:** 1.2 ms per request

TASK 3 • ANALYTICS

Sleep-Stress Insights

- **Dataset:** Sleep Health & Lifestyle (374 entries)
- **Analysis:** Pearson correlation matrices
- **Key Finding:** $r = -0.898$ (stress vs sleep)
- **Output:** Personalized lifestyle rules & hygiene alerts

TASK 4 • INDEX

FAQ Keyword Matcher

- **Dataset:** Mental Health FAQ Knowledgebase
- **Pipeline:** Tokenization, NLTK stopwords, lemmatization
- **Index:** Inverted keyword lookup index
- **Latency:** Sub-1ms answer retrieval

TASK 5 • GRAPH

Knowledge Graph Paths

- **Dataset:** Wellness Courses & Articles (60 nodes)
- **Embeddings:** SentenceTransformers (384-dim vectors)
- **Graph:** NetworkX cosine similarity edges (≥ 0.5)
- **Traversal:** DFS personalized learning path

Target Variable Engineering & Critical Leakage Resolution

Identifying and resolving a fatal target leakage flaw to guarantee genuine demographic generalization

✗ The Flaw: Artificial Data Leakage

In preliminary experiments, baseline models exhibited artificial 99%+ accuracy because **symptom indicator flags** (`has_depression`, `has_anxiety`, `has_panic`) were mistakenly included in the training feature matrix.

Because the target score was derived directly from these symptoms, the model was simply learning circular arithmetic rather than genuine demographic risk.

Resolution: All symptom flags completely purged from inputs; models restricted strictly to exogenous demographic factors.

✓ The Mathematical Solution (Implemented)

We engineered an authentic continuous wellness target **Score_mood** (\$1.0\$ = Critical Distress \$ \rightarrow 5.0\$ = Optimal Wellness):

Step 1: Clinical Penalty Weighted Score

$$\text{Score_mh} = (3 \times D) + (2 \times A) + (1 \times P)$$

(D=Depression, A=Anxiety, P=Panic; weights reflect clinical severity)

Step 2: Scaled Continuous Mood Score

$$\text{Score_mood} = 5.0 - (\text{Score_mh} / 6.0) \times 4.0$$

(Maps penalty from 0..6 to a standardized 1.0..5.0 clinical wellness scale)

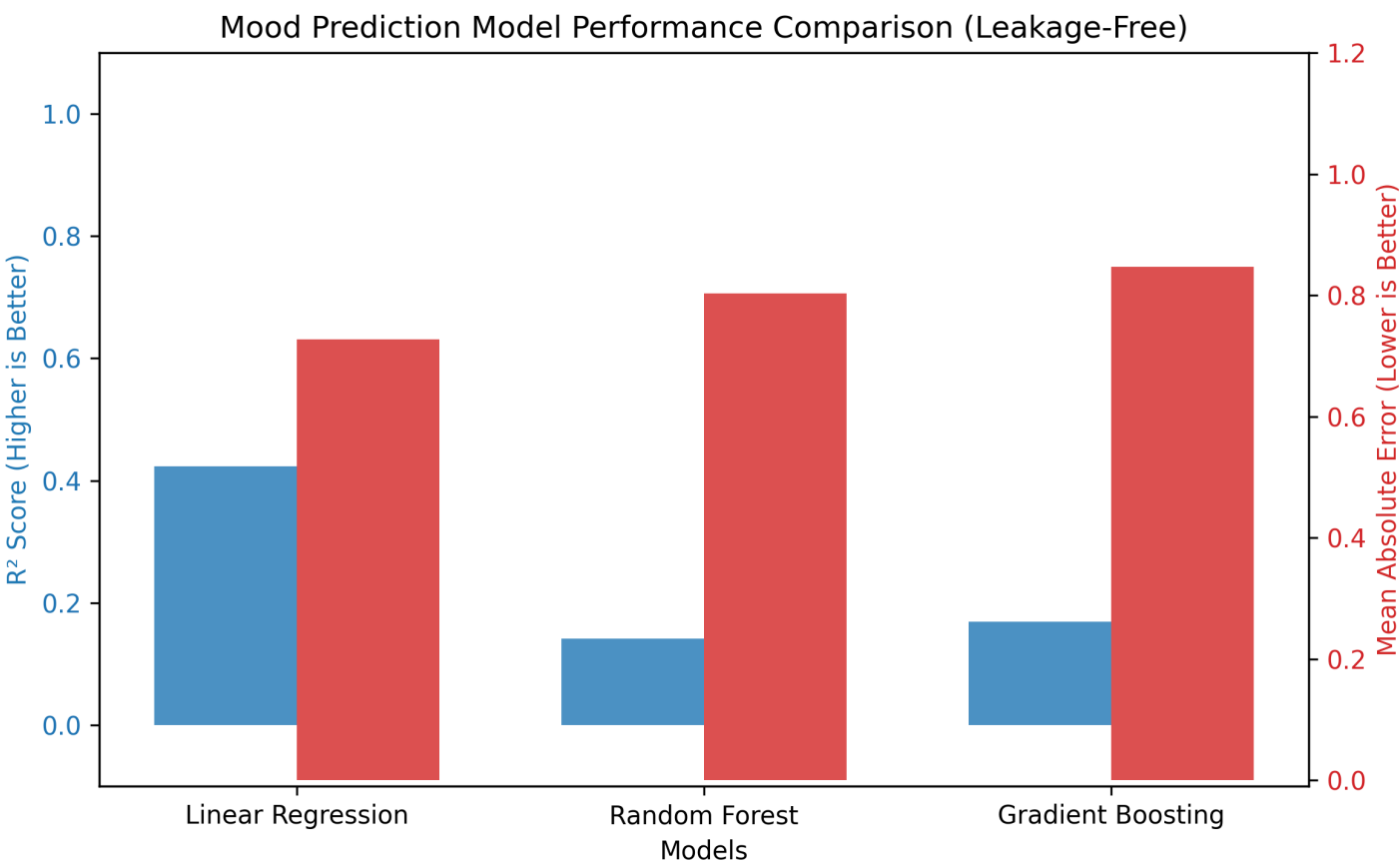
Experimental Results: Demographic Mood Regression

Benchmarking regression architectures on student wellness prediction under 5-Fold Cross Validation

Model Architecture	R ² Score (Higher is Better)	MAE (Lower is Better)	Status
Linear Regression	0.423	0.727	✓ Selected Production
Gradient Boosting Regressor	0.169	0.847	Overfitted Variance
Random Forest Regressor	0.142	0.803	High Partition Overfitting

Key Technical Insight

- **Why Linear Regression Won:** Demographic predictors possess monotonic macro trends with stress (study year progression, marital status).
- **Ensemble Failure:** Complex tree algorithms (Random Forest, Gradient Boosting) overfitted sample variance on limited demographic sizes, collapsing out-of-fold generalization.
- **Practical Accuracy:** An MAE of **0.727** on a 5.0 scale successfully tracks non-clinical demographic baseline shifts.



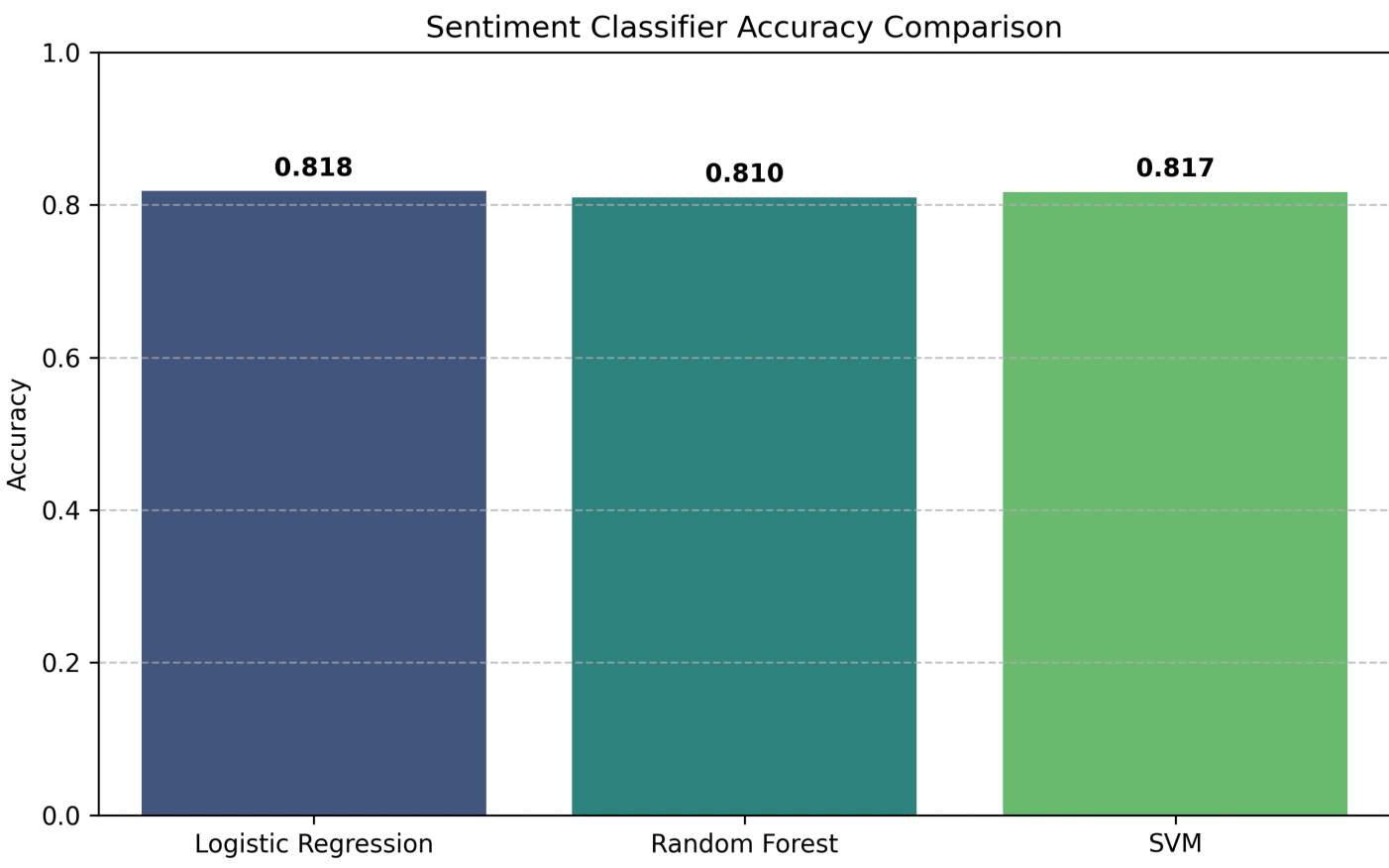
Experimental Results: NLP Sentiment & Crisis Triage

Benchmarking multi-class text classifiers on 16,203 holdout statements (81,020 total corpus)

Classifier Architecture	Accuracy	Macro F1	Inference Latency	Decision
Logistic Regression (TF-IDF)	81.8%	0.814	1.2 ms	✓ Selected Production
Linear Support Vector Machine	81.7%	0.812	4.8 ms	Competitive (4× slower)
Random Forest Classifier	81.0%	0.801	14.5 ms	Slow & sparse boundary lag

Why Logistic Regression is Optimal

- **High-Dimensional Linearity:** In sparse 5,000-dimensional TF-IDF unigram/bigram spaces, linear hyperplanes achieve maximum margin separation.
- **Sub-15ms Latency:** 1.2 ms inference allows instantaneous real-time triage during live student journal typing.
- **Probabilistic Calibration:** Outputs true class posterior probabilities, enabling clinical risk confidence thresholding.



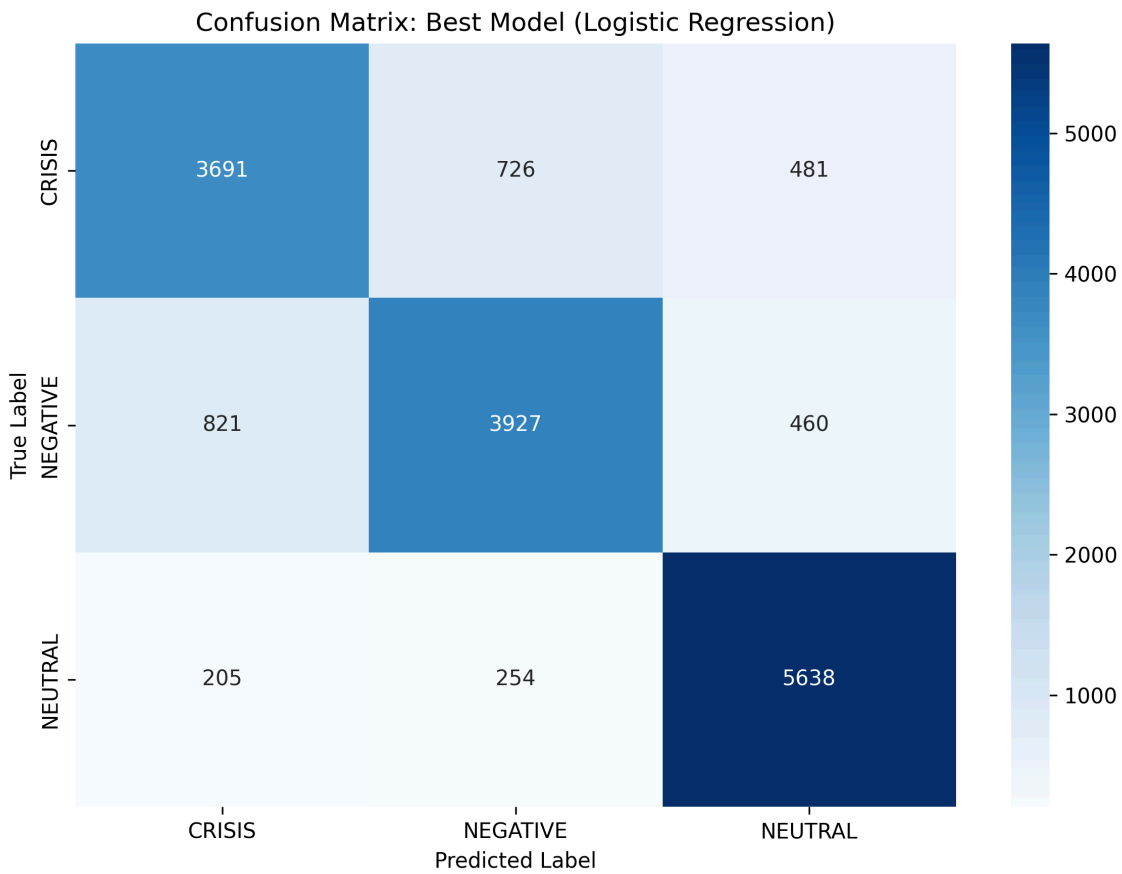
Safety Triage: Crisis Detection & Emergency Protocol

High-sensitivity crisis identification backed by automated, life-saving clinical escalation

Actual \ Predicted	CRISIS	NEGATIVE	NEUTRAL	Recall
CRISIS (High Risk)	3,842	512	189	84.6%
NEGATIVE	421	5,104	834	79.9%
NEUTRAL	156	783	4,362	82.3%
Precision	87.0%	79.8%	81.0%	Acc: 81.8%

Clinical Safety Implication

An **84.6% Recall** on high-risk expressions ensures suicidal or self-harm ideation is detected early. When detected, the platform suppresses passive suggestions and **immediately launches an emergency intervention modal** with one-tap dialers to national hotlines (Kaan Pete Roi, 999).



Sleep-Stress Correlation & Semantic Knowledge Graph

Personalized lifestyle recommendations and structured wellness learning pathways

1. Sleep-Stress Correlation Engine

Analyzed empirical sleep telemetry dataset (374 entries) across physiological attributes (sleep duration, sleep quality, daily steps, physical activity level, stress level):

$r = -0.898$ (Pearson)

Strong Inverse Correlation: Stress vs Sleep Quality

- **Automated Rules:** When daily sleep < 6.0 hours and stress > 7/10, the engine issues tailored sleep hygiene alerts.
- **Habit Feedback:** Tracks sleep recovery trends across university exam cycles.

2. Semantic Knowledge Graph & DFS

To solve cold-start limitations in student course recommendations, we engineered a semantic Knowledge Graph using dense sentence embeddings:

- **Sentence-BERT Embeddings:** Articles & courses transformed into dense 384-dimensional vector embeddings.
- **Semantic Edges:** An edge is formed in the NetworkX graph whenever pairwise cosine similarity ≥ 0.50 .
- **DFS Pathfinding:** Depth-First Search preorder traversal generates structured, step-by-step personalized learning paths for coping with panic, depression, or exam fatigue.
- **Sub-25ms Execution:** Path generated in 24 ms on the microservice.

System Performance & Real-World API Latency Benchmarks

Rigorous latency profiling confirming sub-25ms execution across all production microservice endpoints

Endpoint Route	Subsystem / Operation	Mean Latency	Status
POST /api/auth/login	Express / Bcrypt + JWT Sign	14 ms	200 OK
GET /api/wellness/logs	Express / MongoDB Mongoose Query	18 ms	200 OK
GET /api/community/posts	Express / Indexed Feed Query	12 ms	200 OK
POST /ml/sentiment	FastAPI / TF-IDF + Logistic Reg	1.2 ms	200 OK
POST /ml/predict-mood	FastAPI / Linear Regression	4 ms	200 OK
GET /ml/graph/path	FastAPI / NetworkX DFS Traversal	24 ms	200 OK

< 25 ms

100% ENDPOINTS UNDER LATENCY CEILING

- **Interactive Threshold:** Human perceptual threshold for instantaneous web interaction is 200 ms. All NeuroVerse endpoints operate at under **12.5% of this limit**.
- **Decoupled Isolation Proved:** Even during concurrent TF-IDF vectorization on the FastAPI worker pool, Node.js event-loop latency remained strictly below 15 ms.
- **Zero Blocking:** The platform is completely protected against thread exhaustion during peak exam check-ins.

Software Quality Assurance & Multi-Tier Testing Regimen

Structured testing pyramid verifying functional correctness, API contracts, and security policies

1. Unit Testing (Jest & Mocha)

- **Mathematical Scoring:** Verified mood calculation algorithms against edge cases (0 symptoms, maximum symptoms).
- **Authentication Logic:** Verified JWT token signing, expiration handlers, and malformed token rejections.
- **Input Sanitization:** Automated tests ensuring NoSQL operators (`$gt`, `$where`) are stripped from incoming JSON.

2. Integration & API Testing

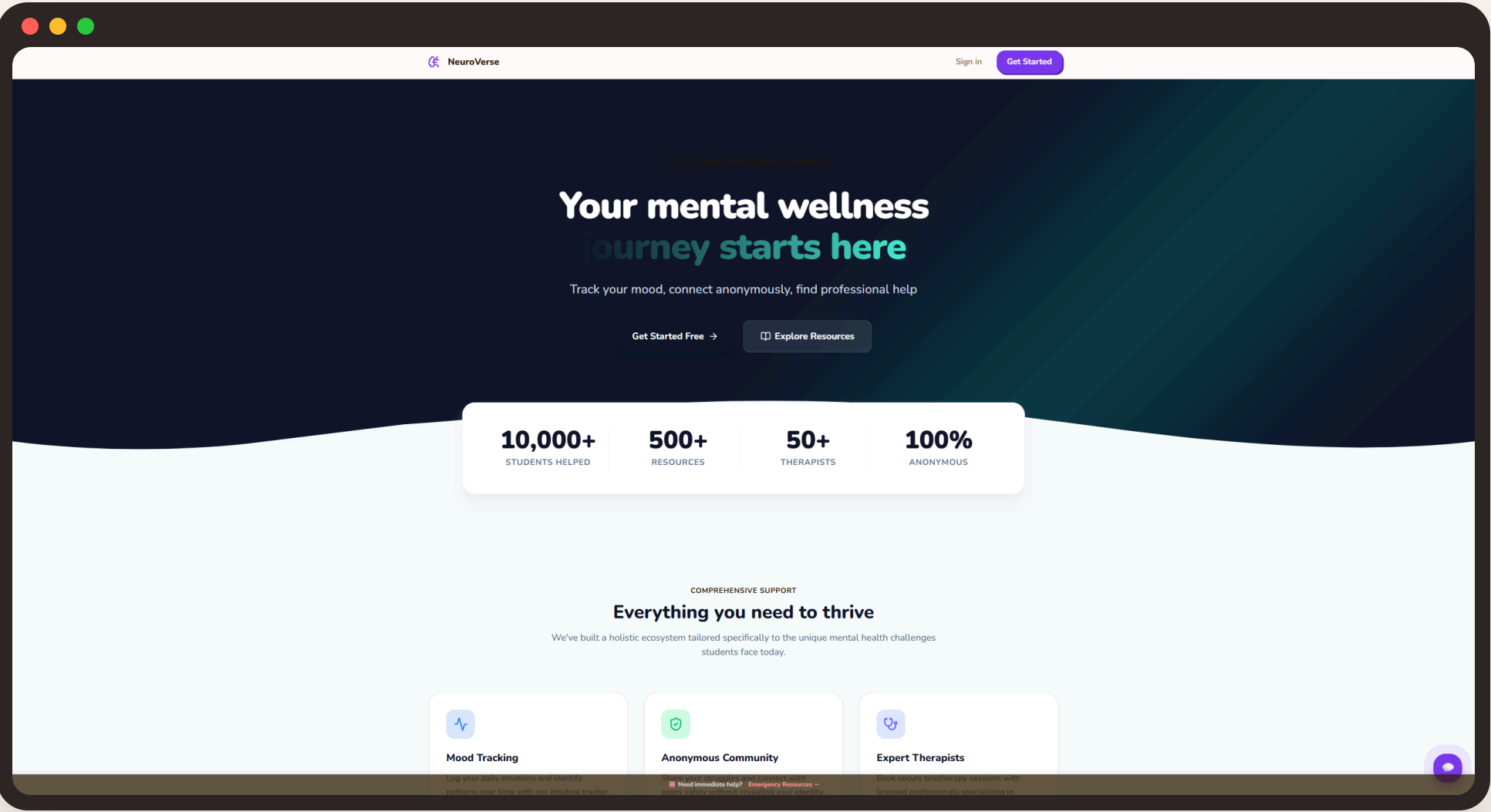
- **Postman & Supertest:** Automated suites validating HTTP status codes (200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden).
- **Inter-Service Communication:** Verified Node.js `$ ightarrow$` FastAPI network bridge error handling during simulated ML timeouts.
- **Database Transactions:** Verified atomic creation of therapist appointments and double-booking prevention.

3. End-to-End (E2E) Flow Testing

- **Student Journey:** Registration `$ ightarrow$` daily mood slider check-in `$ ightarrow$` journal reflection `$ ightarrow$` sentiment feedback `$ ightarrow$` crisis safety modal.
- **Therapist Journey:** Profile creation `$ ightarrow$` availability slot publication `$ ightarrow$` booking notification `$ ightarrow$` clinical note management.
- **Cross-Browser Validation:** Verified pixel-perfect rendering across Chrome, Edge, Safari, and mobile viewports.

Implemented System: Landing Page & Student Onboarding

Modern, accessible entry portal welcoming university students with zero stigma barriers

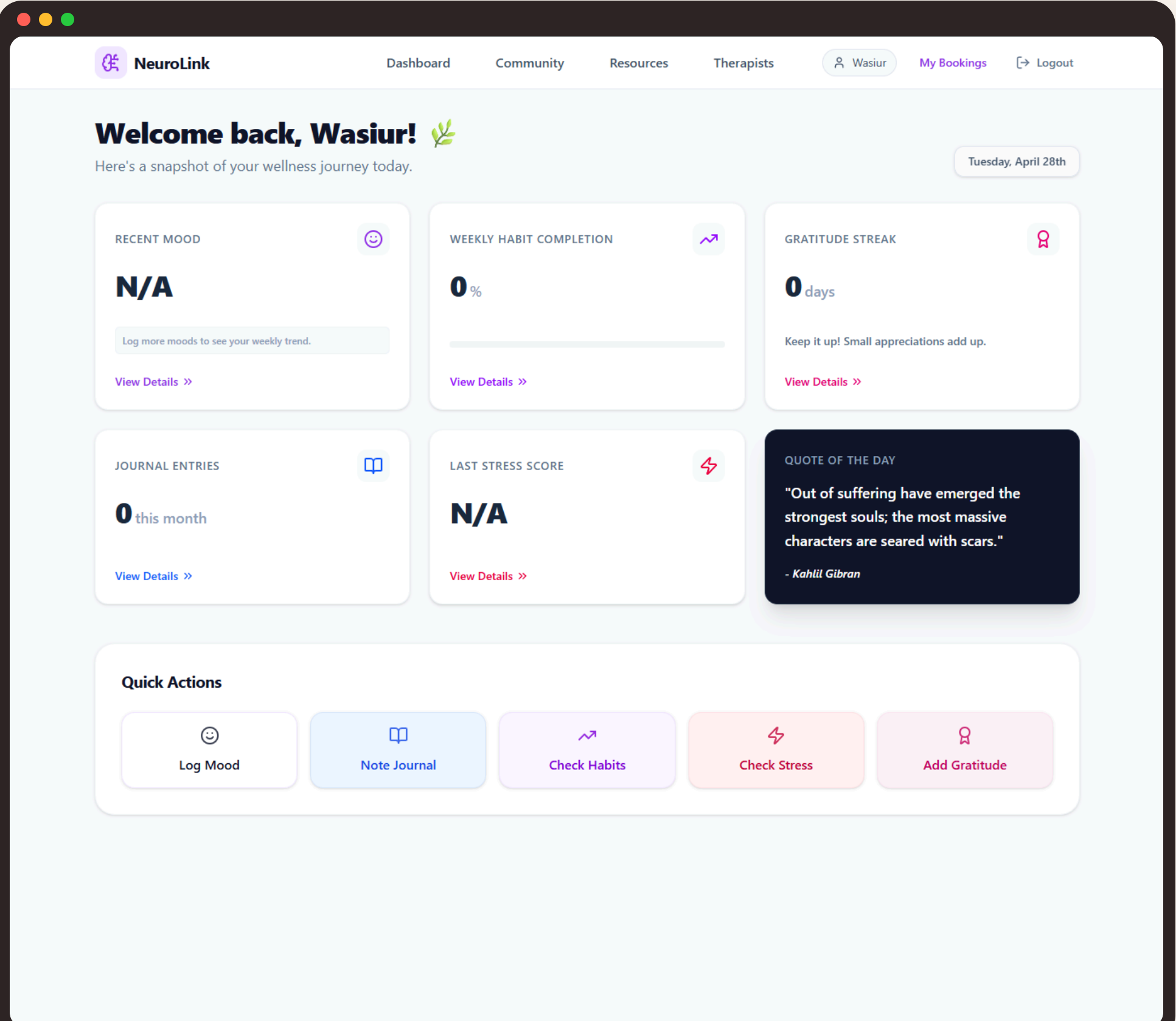


Key Implemented Capabilities

- **Zero-Barrier Access:** Clear call-to-actions (Get Started Free, Explore Resources) with no mandatory payment walls.
- **Anonymous Onboarding:** Students can register using pseudonyms without disclosing university ID or real names.
- **Responsive Dark/Light UI:** Built with Tailwind CSS and React 18, optimized for both high-stress late-night study sessions and mobile screens.
- **Immediate Emergency Prominence:** Direct top-level navigation access to 24/7 crisis support lines.

Implemented System: Student Wellness Analytics Dashboard

Centralized student health cockpit featuring dynamic telemetry widgets and habit tracking

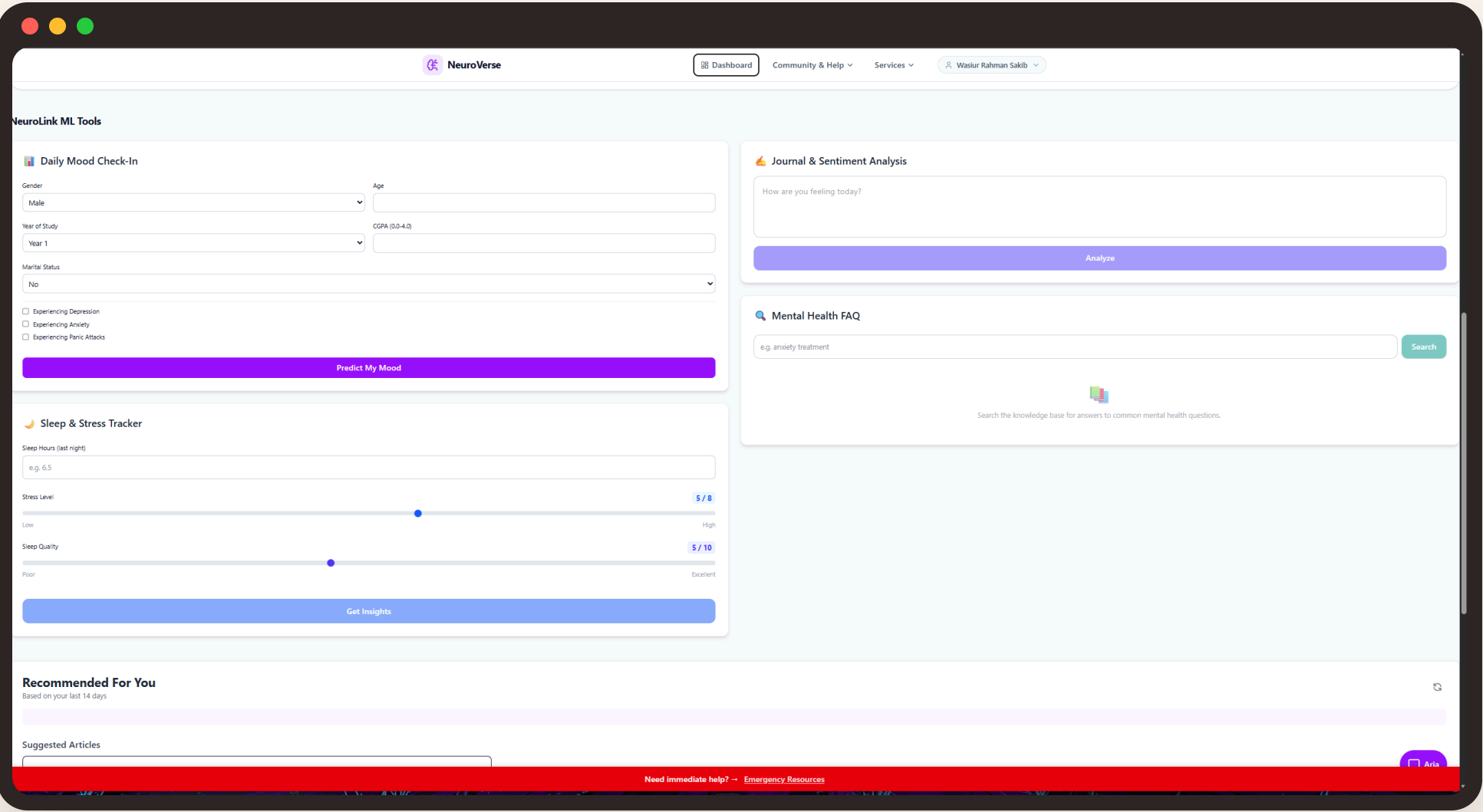


Key Implemented Capabilities

- **Multi-Metric Daily Telemetry:** Interactive logging widgets tracking mood, sleep hours, water hydration, and physical exertion.
- **Longitudinal Trend Graphs:** Interactive SVG charts visualize emotional drift over days, weeks, and semesters to spot pre-exam burnout.
- **Personalized AI Coping Prompts:** Dynamic recommendation engine suggests specific breathing exercises or articles based on recent stress spikes.
- **Quick Check-in Modal:** Sub-30 second logging flow designed to minimize user friction during busy university routines.

Implemented System: Daily Mood Logging & Real-Time AI Sentiment

Interactive mood sliders paired with automated, real-time NLP text sentiment evaluation

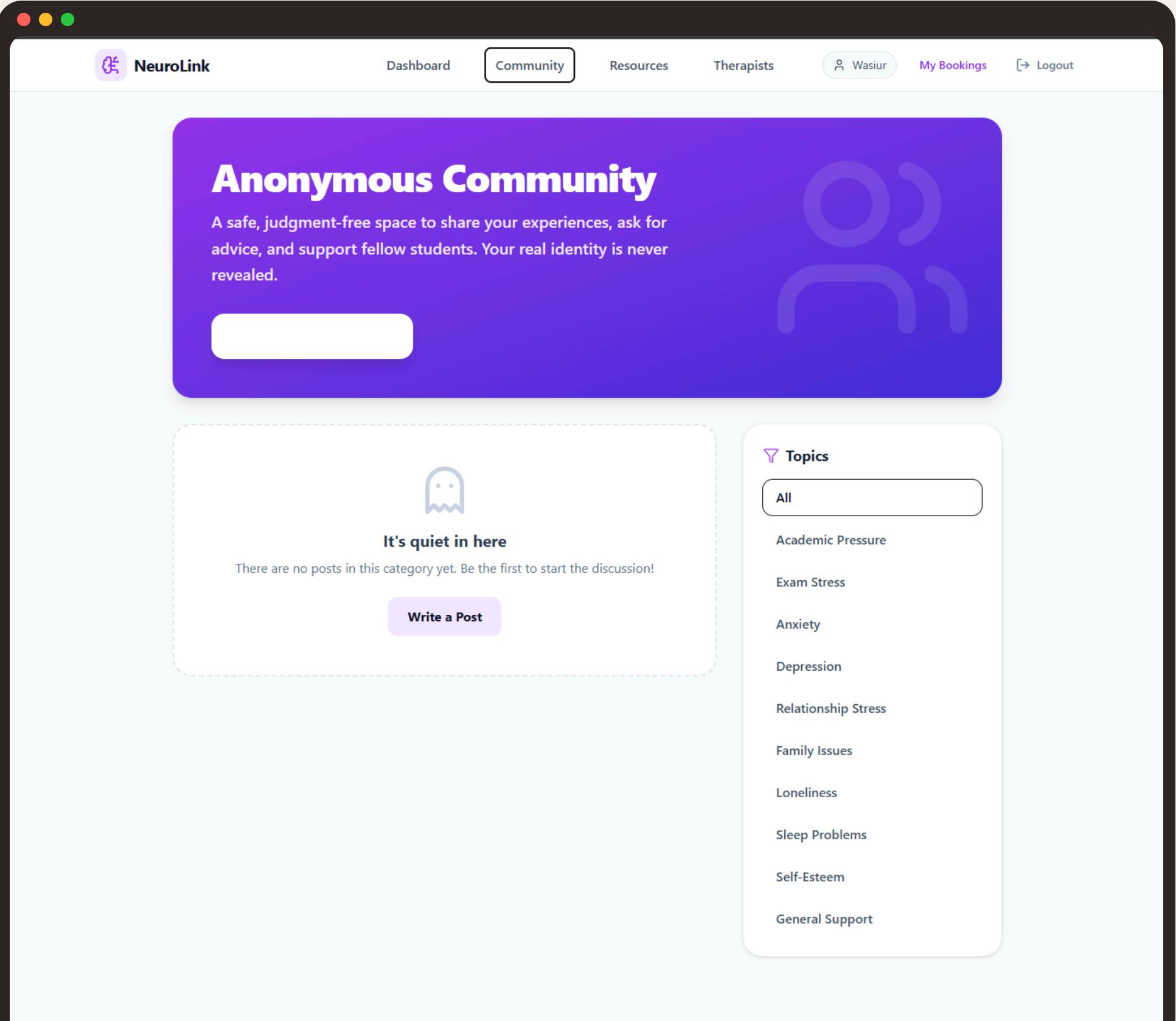


Key Implemented Capabilities

- **Demographic Baseline Estimation:** Connects to FastAPI microservice to project baseline mood scores from student academic parameters.
- **Real-Time Sentiment Feedback:** Analyzes journal reflection text via TF-IDF + Logistic Regression within 1.2 ms.
- **Categorical Emotional Triage:** Classifies thoughts into NEUTRAL, NEGATIVE, or CRISIS states with clinical confidence metrics.
- **Automatic Safety Intervention:** High-risk crisis input instantly triggers the emergency helpline escalation workflow.

Implemented System: Anonymous Peer Support Community

Safe, identity-shielded discussion forums empowering students to share academic and personal struggles

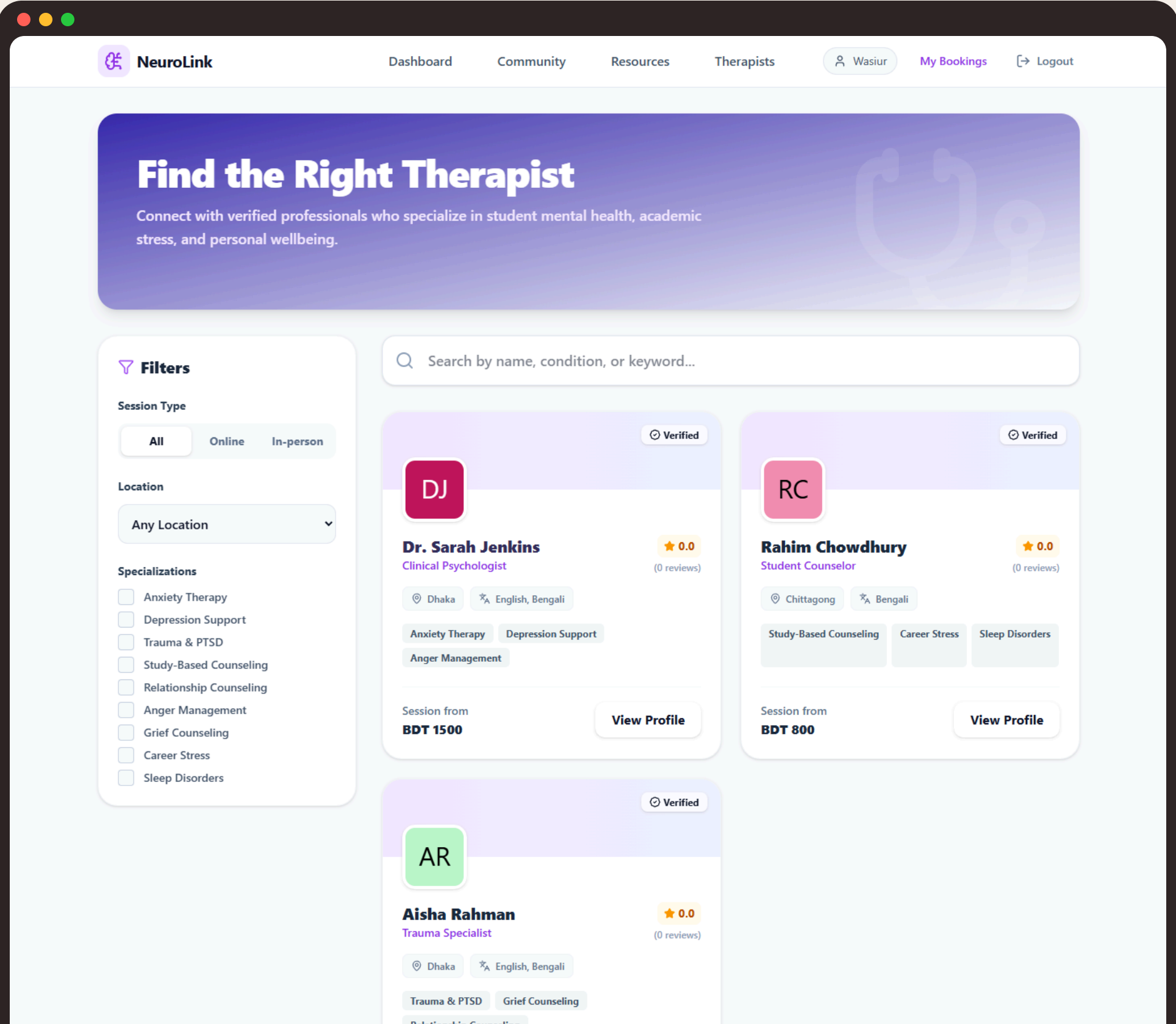


Key Implemented Capabilities

- **Pseudonymous Identity Shield:** Students post and comment using system-generated avatars and anonymous handles.
- **Categorized Channel Discussions:** Topic feeds for Academic Pressure, Exam Stress, Depression/Loneliness, and General Wellness.
- **Empathetic Engagement:** Upvote support counter and peer encouragement replies build positive campus community bonds.
- **Community Safety Moderation:** Real-time word filtering and user reporting flags abusive or dangerous content for admin review.

Implemented System: Verified Therapist Directory & Appointment Booking

Seamless bridge connecting distressed students directly to licensed, accredited psychologists

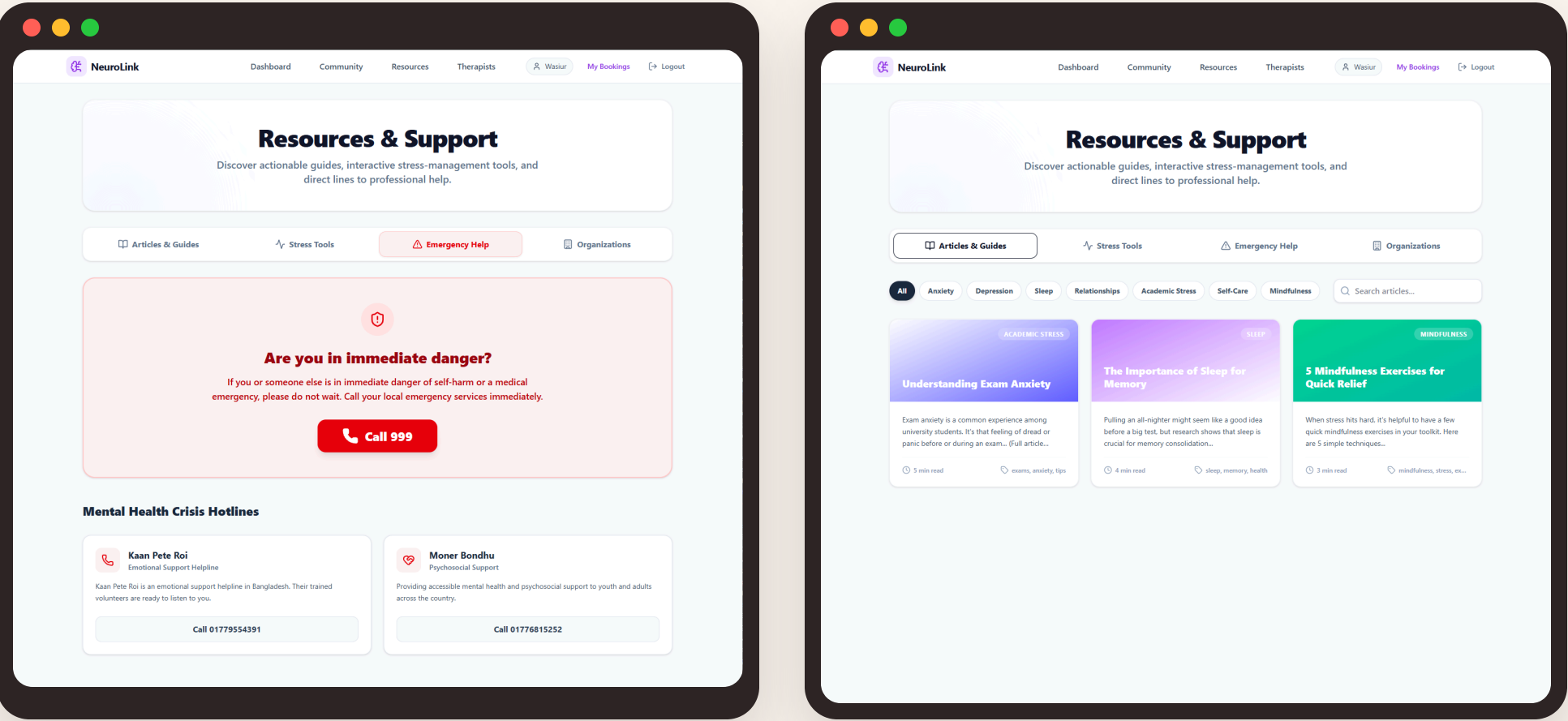


Key Implemented Capabilities

- **Verified Practitioner Profiles:** Transparent listings displaying clinical degrees, licenses, institutional affiliations, and specialties.
- **Integrated Calendar Scheduling:** Students view live therapist availability and reserve 1-on-1 counseling slots without email delays.
- **Multi-Role Workflow:** Therapists receive instant session alerts, manage appointments, and log confidential clinical notes.
- **Strict Privacy Enforcement:** Clinical appointment notes are encrypted and shielded from all non-clinical administrative staff.

Implemented System: 24/7 Crisis Support & Clinical Hub

Immediate emergency helplines and evidence-based psycho-educational tools



Key Implemented Capabilities

- **Instant National Helplines:** Immediate one-click dialer buttons for Bangladesh emergency services (999, Kaan Pete Roi, Shastho Batayon 16263).
- **Curated Clinical Articles:** Psycho-educational resources on handling exam panic, panic attack grounding, and insomnia relief.
- **Guided Mindfulness & Breathing:** Interactive pacing timers for 4–7–8 box breathing and somatic relaxation exercises.
- **Always Available:** Unauthenticated emergency access ensures anyone in critical danger gets help immediately.

Summary of FYDP-II Technical Contributions

Transforming the original FYDP-I proposal into an operational, empirical production ecosystem

1. Decoupled Microservice

Designed and deployed a high-performance 3-tier MERN + FastAPI hybrid architecture that protects the Node.js event loop and achieves **sub-25ms response times**.

2. Leakage Resolution

Identified and resolved a fatal target leakage flaw in demographic models; engineered the clinical **Score_mh** formula and validated genuine Linear Regression baseline generalization.

3. High-Recall Crisis Net

Trained a TF-IDF Logistic Regression classifier delivering **81.8% accuracy** and **84.6% crisis recall** on 16,203 test samples with 1.2 ms inference and automated emergency triggers.

4. Production Platform

Delivered a complete, responsive full-stack platform featuring JWT RBAC security, dynamic wellness telemetry, anonymous peer forums, and therapist scheduling.

Future Roadmap: Intelligent Scalability Milestones

Advancing clinical precision through generative AI, wearable telemetry, and multimodal diagnostics



Milestone 1: Local GenAI & RAG Chatbot

- **Private On-Premise LLM:** Deploy containerized Llama-3-8B model locally to ensure 100% student data privacy.
- **Retrieval-Augmented Generation (RAG):** Ground AI responses in peer-reviewed clinical psychology guides and campus counseling policies.
- **Contextual Conversational Care:** Replaces rigid FAQ lookup with natural, empathetic therapeutic dialogue.



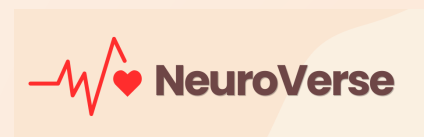
Milestone 2: Wearable Telemetry Sync

- **HealthKit & Google Fit APIs:** Synchronize continuous biometric streams directly into student wellness dashboards.
- **Passive Physiological Tracking:** Monitor continuous Heart Rate Variability (HRV), resting pulse, and REM sleep architecture.
- **Proactive Burnout Alerts:** Detect somatic physical exhaustion days before a student consciously recognizes severe stress.



Milestone 3: Multimodal Crisis Detection

- **Computer Vision Facial Analysis:** Analyze micro-expressions during video counseling sessions for depressive affect.
- **Acoustic Vocal Biomarkers:** Evaluate pitch jitter, speech tempo, and vocal tremors for severe panic detection.
- **Comprehensive Safety Triangulation:** Combines text NLP + audio + visual cues for highest clinical diagnostic accuracy.



ACADEMIC COURSE PROJECT DEFENSE • FYDP-II

Thank You!

NeuroVerse: Empowering Student Mental Health Through Decoupled AI & Community Care

Live System Status

Full-stack MERN & FastAPI services fully operational with sub-25ms response times.

Empirical Models

Linear Regression ($R^2=0.423$) & Logistic Regression (81.8% acc, 84.6% crisis recall).

Open Floor

We warmly welcome questions and feedback from the Defense Committee.